NTIRE 2026 Image Super-Resolution (×4) Challenge Factsheet
-title of the contribution-

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1. Introduction

This factsheet template is meant to structure the description of the contributions made by each participating team in the NTIRE 2026 challenge on image super-resolution (×4).

Ideally, all the aspects enumerated below should be addressed. The provided information, the codes/executables, and the achieved performance on the testing data are used to decide the awardees of the NTIRE 2026 challenge.

Reproducibility is a must and needs to be checked for the final test results in order to qualify for the NTIRE awards.

The main winners will be decided based on overall performance and a number of awards will go to novel, interesting solutions and to solutions that stand up as the best in a particular subcategory the judging committee will decide. Please check the competition webpage and forums for more details.

The winners, the awardees and the top-ranking teams will be invited to co-author the NTIRE 2026 challenge report and to submit papers with their solutions to the NTIRE 2026 workshop. Detailed descriptions are appreciated.

The factsheet, source codes/executables, trained models should be sent to **all of the NTIRE 2026 challenge organizers (Zheng Chen, Kai Liu, Jingkai Wang, Jue Gong, Jiatong Li, Radu Timofte, and Yulun Zhang)** by email.

2. Email final submission guide

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cc: your_team_members
Title: NTIRE 2026 Image Super-Resolution (×4) Challenge - TEAM_NAME - TEAM_ID

To get your TEAM_ID, please register at Google Sheet. Please fill in your Team Name, Contact Person, and Contact Email in the first empty row from the top of the sheet. Body contents should include:

a) team name
b) team leader's name and email address
c) rest of the team members
d) user names on NTIRE 2026 Codabench competitions
e) Code, pre-trained model, and factsheet download command, e.g. `git clone ...`, `wget ...`
f) Result download command, e.g. `wget ...`
   - Please provide different URLs in e) and f)

Factsheet must be a compiled pdf file together with a zip with .tex factsheet source files. Please provide a detailed explanation.

3. Code Submission

The code and trained models should be organized according to the GitHub repository. This code repository provides the basis for comparing the various methods in the challenge. **Code scripts based on other repositories will not be accepted.** Specifically, you should follow the steps below.

1. Git clone the repository
2. Put your model script under the `models/[Team_ID]_[Model_Name]` folder
3. Put your pretrained model under the `model_zoo/[Team_ID]_[Model_Name]` folder

4. Modify `model_path` in `test.py`. Modify the imported models
5. `python test.py` (restore images, details in GitHub)
6. `python eval.py` (eval results, details in GitHub)

Please send us the command to download your code, e.g. `git clone [Your repository link]` When submitting the code, please remove the input and output images in the (any) `data` folder to save the bandwidth.

4. Factsheet Information

The factsheet should contain the following information. Most importantly, you should describe your method in detail. The training strategy (optimization method, learning rate schedule, and other parameters such as batch size, and patch size) and training data (information about the additional training data) should also be explained in detail.

4.1. Team details

- Team name: AIT
- Team leader name: Yang Ji
- Team leader address: Hangzhou, Zhejiang Province, China
  phone number: 18756164789
  email: jiyang@kunbyte.com
- Rest of the team members: Zonghao Chen, Zhihao Xue, Junqin Hu
- Team website URL: None
- Affiliation: KUNBYTE
- User names: jiyang
- Best scoring: 22.79
- Link to the codes:
  https://github.com/jiyang0315/NTIRE2026_ImageSR_x4.git

4.2. Method details

4.2.1. Overview

The overall framework, as shown in Figure 1, utilizes a diffusion model for high-resolution image restoration. The process begins by extracting semantic information and degradation features from the low-resolution (LR) image. The degradation information is captured by the degradation descriptor extractor, which, after passing through an MLP adapter, generates the degradation token. Simultaneously, the structure map (edge information) is extracted, providing crucial spatial structure for the restoration process.

The extracted semantic tokens, degradation tokens, and structure map are then fused as joint conditions and fed into the diffusion denoiser, where cross-attention is applied to iteratively refine the image from noise to high-resolution output.

During training, spatially asymmetric noise injection is used and compared with standard uniform noise, improving the denoising and image restoration performance.

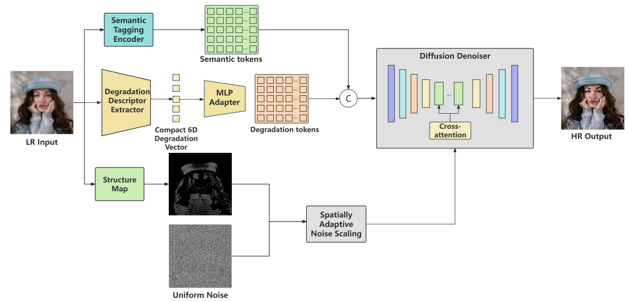

Figure 1. Overview of the proposed framework.

This framework combines degradation information, semantic understanding, and structural preservation to effectively recover high-resolution images with rich details.

4.2.2. Network Design

Our model has three key components:

(1) Degradation-aware token injection. The model begins by extracting semantic cues from the input low-resolution (LR) image using the Semantic Tagging Encoder. Alongside this, a Degradation Descriptor Extractor captures six degradation features of the image, including blur level, noise level, JPEG blocking intensity, edge density, brightness, and contrast. These features are then projected into a compact 6D degradation vector using a MLP Adapter. The degradation vector is concatenated with the semantic tokens before being processed through the cross-attention mechanism in the Diffusion Denoiser.

(2) Dynamic degradation token. To enhance the conditioning at different stages of the denoising process, the degradation token can be updated with diffusion timestep information. This allows for stronger structural guidance during the early denoising steps and finer texture details in the later stages of the process.

(3) Spatial asymmetric noise injection. Instead of using uniform noise, we use an edge map derived from the LR image to scale the noise. This results in spatially adaptive noise scaling, where edge-rich regions receive less noise perturbation, while flat areas maintain standard noise strength. This strategy helps preserve the image's geometric structure during diffusion training, improving the quality of the restored high-resolution output.

4.2.3. Training Strategy

Training and testing details. We use LSDIR as the training set. Training is performed at resolution 512 with mixed precision (fp16). We use AdamW with a fixed learning rate of 5×10^{-5}. The batch size is set to 96. Checkpoints are saved every 10k iterations. For training options, we enable degradation-token conditioning and dynamic token modulation; for structure-aware training, we use Sobel-based spatial noise modulation with $\alpha = 0.6$. For inference, we use

50 diffusion steps, guidance scale 5.5, and $\times 4$ upscaling with tiled inference.

Experimental setup and comparison. We evaluate on the RealSR benchmark and compare with ResShift[6], StableSR[2], OSEDiff[3], SeeSR[4], PASD[5], PiSA-SR[1], InvSR[7]. We report eight metrics: PSNR↑, SSIM↑, LPIPS↓, NIQE↓, NRQM↑, PI↓, CLIP-IQA↑, and MUSIQ↑.The evaluation results are shown in the following table, from which we can see that our model is better than these two models in most indicators.

Experimental Results. The experimental results demonstrate that our algorithm achieves superior performance compared to other methods, particularly in the NIQE, PI, and MUSIQ metrics. Our approach consistently yields higher scores in these evaluation criteria, indicating better image quality, preservation of perceptual details, and improved alignment with human visual perception. These results highlight the effectiveness of our method in enhancing image restoration performance across multiple evaluation metrics.

Table 1. Quantitative comparison on RealSR. ↑ indicates higher is better and ↓ indicates lower is better.

Method	PSNR↑	SSIM↑	LPIPS↓	NIQE↓	NRQM↑	PI↓	CLIP-IQA↑	MUSIQ↑
ResShift[6]	25.453	0.725	0.266	7.349	6.202	5.55	0.585	56.189
StableSR[2]	19.536	0.615	0.394	7.174	3.6	6.826	0.519	38.485
OSEDiff[3]	25.254	0.736	0.23	4.342	6.532	3.969	0.677	67.579
SeeSR[4]	23.848	0.692	0.255	3.872	6.635	3.666	**0.701**	68.558
PASD[5]	**26.792**	**0.762**	0.215	4.867	5.546	4.77	0.521	58.689
PiSA-SR[1]	25.287	0.749	**0.208**	4.388	6.421	4.033	0.668	67.995
InvSR[7]	24.928	0.734	0.211	4.75	**6.672**	4.06	0.671	66.371
Ours	23.16	0.662	0.284	**3.622**	6.619	**3.531**	0.657	**68.667**

References

[1] Lingchen Sun, Rongyuan Wu, Zhiyuan Ma, Shuaizheng Liu, Qiaosi Yi, and Lei Zhang. Pixel-level and semantic-level adjustable super-resolution: A dual-lora approach. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2333–2343, 2025. 3

[2] Jianyi Wang, Zongsheng Yue, Shangchen Zhou, Kelvin CK Chan, and Chen Change Loy. Exploiting diffusion prior for real-world image super-resolution. *International Journal of Computer Vision*, 132(12):5929–5949, 2024. 3

[3] Rongyuan Wu, Lingchen Sun, Zhiyuan Ma, and Lei Zhang. One-step effective diffusion network for real-world image super-resolution. *Advances in Neural Information Processing Systems*, 37:92529–92553, 2024. 3

[4] Rongyuan Wu, Tao Yang, Lingchen Sun, Zhengqiang Zhang, Shuai Li, and Lei Zhang. Seesr: Towards semantics-aware real-world image super-resolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 25456–25467, 2024. 3

[5] Tao Yang, Rongyuan Wu, Peiran Ren, Xuansong Xie, and Lei Zhang. Pixel-aware stable diffusion for realistic image super-resolution and personalized stylization. In *European conference on computer vision*, pages 74–91. Springer, 2024. 3

[6] Zongsheng Yue, Jianyi Wang, and Chen Change Loy. Resshift: Efficient diffusion model for image super-resolution by residual shifting. *Advances in neural information processing systems*, 36:13294–13307, 2023. 3

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